1; Explain behaviour of hub device

Ans: Device A is sending data to Device B through hub, so source mac is A and destination mac is B. Data goes from device A to hub and hub send data to connected devices B and device C: When data goes to device C it see dest mac so it is not match dest mac B. Device C cannot open data. When data from device A goes to device B through hub, device B opens the data because destination mac is B. B device reply source mac is B and destmac is A. So data goes from device B to device A and device C through hub. Device C cannot open data because destination mac is A. The drawback in hub is more traffic and collision domain.

2. Explain behaviour of switch device

2. Ans: Data is sending from device A to device B through switch. When data comes from device A to switch, it see which port data comes from and source mac and note it on mac table so it is called address learning. After address learning, switch see the destination mac B on mac table if it is not available switch broadcast the data to device B and device C. Device C cannot open data because destination mac is B. B device send replying to A device, source mac is B and destination mac is A. B device send data to switch through port no 2 to device A. Switch ~~note~~ see the port no and source mac and note it on mac table so it is called address learning. After address learning, switch see the destination mac A on mac table it is available and connected to port no 1. So switch acts as a Unicast device. The only drawback is broadcast traffic still occur.

3. Explain behaviour of router device

3.) Behaviour of router device:

HR department PC (10.0.0.10) is sending data to production department PC (192.168.1.10).

HR department PC (10.0.0.10) data goes to switch and switch uses address learning it see destination mac entry on mac table if not available it broadcast data and if available it unicast data. Packet comes from switch to router through router interface e0. When the HR department PC (10.0.0.10) data comes to router, it will see destination IP route on routing table. Routing table contains route 192.168.1.0/24 – e3. So router send data to router interface e3 and e3 is connected to switch. Switch uses the address learning so it see destination mac entry on mac if not available it broadcast data and if mac entry available it unicast data. Production department PC (192.108.1.10) replying source IP is 192.108.1.10 and destination IP is 10.0.0.10. Production department PC (192.108.1.10) send the data to switch and switch uses address learning and send data to router through e3 and router see destination ip route on routing table it contains 10.0.0.0/8 – e0 so data goes from router to switch through e0 and switch uses address learning and if mac entry is not available it broadcast data or if available it ~~send~~ unicast data; so switch send data to HR department PC 10.0.0.10